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CSD-380

Assignment 2.2

In Chapter 6 of The DevOps Handbook, the authors present a case study that examines challenges faced by LinkedIn in 2011 and the initiative led by their VP of Engineering to address mounting technical problems.

LinkedIn initially ran on a monolithic Java application called Leo. Leo used servlets and JDBC connections to manage LinkedIn's backend databases and served as the foundation for much of the site's functionality. It likely handled features such as account creation and login, user profiles, connections, messaging, search functions and job postings. As traffic increased, LinkedIn decoupled two important services from Leo: member search and the member connection graph. The decoupling of these services was an attempt to reduce the load on Leo. Instead of continuing to expand the monolithic application, these functions were removed and rebuilt as independent services that could scale separately.

Despite these changes, Leo continued to become a bottleneck as LinkedIn grew. By 2010, nearly one hundred services were operating outside of Leo, while Leo itself was only being deployed once every two weeks. The application had become increasingly difficult to troubleshoot and modify safely. As a result, LinkedIn accumulated significant technical debt while continuing to add new features and services.

Kevin Scott, LinkedIn's VP of Engineering in 2011, recognized that the company had outgrown its original architecture. Although many services had been separated from Leo, they still depended on the monolithic application. To address these problems, Scott launched an initiative called Operation InVersion. The initiative required engineers to temporarily stop developing new features and instead focus on improving the underlying architecture. A major goal was to gradually break Leo into smaller, more manageable, stateless services that could be deployed and scaled independently.

The primary lesson from Operation InVersion is that organizations cannot focus exclusively on delivering new features while neglecting the health of their underlying systems. LinkedIn's rapid growth exposed weaknesses in its architecture. Rather than continuing to build on an increasingly fragile foundation, the company chose to invest in long-term improvements to reliability and scalability. The success of Operation InVersion demonstrated the importance of periodically prioritizing infrastructure and technical debt reduction alongside feature development. This case study illustrates a core DevOps principle: sustainable growth requires continuous investment in the systems and tools that support software delivery.

References

Kim, G., Humble, J., Debois, P., Willis, J., & Allspaw, J. (2021). *The DevOps handbook: How to create world-class agility, reliability, and security in technology organizations* (2nd ed.). IT Revolution Press.